Introduction to Data Science (BE040AT011)

SOCIAL MEDIA USAGE ANALYSIS AMONG STUDENTS

(*Problem Based Learning (PBL)*)

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CERTIFICATE

This is to certify that project work embodied in this report entitled **project name** was carried out by **studentname1 (Enrollment no), studentname2 (Enrollment no)** at **C. K. Pithawala College of Engineering and Technology, Surat** as a part of **Problem Based Learning (PBL)** for subject **Introduction to Data Science (BE040AT011)**. This project work has been carried out under my supervision.

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Place : Surat, Gujarat

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DECLARATION

We hereby certify that We are the sole author of this report and that neither any part of this work nor the whole of the work has been submitted for a degree to any other University or Institution.

We declare that this is a true copy of our report, including any final revisions, as approved by my supervisor.

Date :

Place : Surat,Gujarat

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Abstract

This mini project focuses on analyzing social media usage among students and its impact on academic performance, sleep patterns, and mental health. The dataset consists of various attributes such as daily usage hours, most used platform, sleep duration, mental health score, and addiction level. The data was analyzed using Python libraries including NumPy and Pandas for data handling, and Matplotlib and Seaborn for visualization. Statistical analysis and exploratory data analysis (EDA) were performed to identify trends, patterns, and relationships between variables. The results provide insights into how social media usage influences students' academic and personal well-being, highlighting the importance of maintaining a balanced digital lifestyle.

1 Project Introduction

Data Science plays a significant role in extracting meaningful insights from raw data and supporting data-driven decision-making. In today’s digital era, social media has become an essential part of students’ daily lives, influencing their habits, productivity, and overall well-being. Excessive usage of social media platforms can affect academic performance, sleep patterns, and mental health, making it an important area of study.

This project focuses on analyzing social media usage among students and understanding its impact on various factors such as academic performance, mental health, and daily routines. By applying data science techniques, the project aims to identify patterns and relationships between screen time and student behavior. The analysis is carried out using Python-based tools and libraries, enabling efficient data processing, statistical analysis, and visualization.

The insights derived from this study highlight the importance of maintaining a balanced lifestyle and demonstrate how data-driven approaches can be used to understand and improve student well-being.

2 Problem Definition & Objective

Problem Definition:

The primary problem addressed in this project is to analyze the impact of social media usage on students' academic performance, mental health, and daily habits such as sleep patterns. With the rapid growth of digital platforms, students are increasingly spending more time on social media, which may influence their productivity, well-being, and overall lifestyle. Therefore, it is important to study these effects using a data-driven approach to understand the relationships between usage patterns and student outcomes.

Objective:

The objective of this project is to examine social media usage among students and identify how it affects key aspects such as academic performance, mental health, and sleep duration. The project also aims to apply statistical analysis and exploratory data analysis techniques to uncover meaningful patterns and correlations within the dataset. Additionally, feature engineering methods are used to enhance the dataset, and visualization techniques are employed to present the findings in a clear and interpretable manner. The overall goal is to derive insights that can help promote a balanced and healthy use of social media among students.

3 Dataset Description

The dataset used in this project is taken from Kaggle represents social media usage patterns among students. It consists of a total of 101 records, with each record corresponding to an individual student. The dataset contains 13 attributes, including Student ID, Age, Gender, Academic Level, and Country, which provide demographic information about the students.

In addition to basic details, the dataset includes behavioral and analytical attributes such as Average Daily Usage Hours and Most Used Platform, which help in understanding students' social media habits. It also contains impact-related features such as Affects Academic Performance, Sleep Hours per Night, Mental Health Score, Conflicts Over Social Media, and Addiction Score. These attributes are used to analyze how social media usage influences students' academic outcomes, mental well-being, and lifestyle patterns.

This dataset was selected due to the increasing relevance of social media usage among students and its potential effects on their academic and personal lives. The structured nature of the dataset makes it suitable for performing statistical analysis, exploratory data analysis, and visualization to derive meaningful insights.

4 Tools and Technologies Used

This project was implemented using Python 3 as the primary programming language due to its simplicity and powerful capabilities for data analysis. The development and execution were carried out in the Google Colab environment, which provides an interactive platform for writing and running Python code efficiently.

Several Python libraries were utilized to perform different tasks throughout the project. NumPy was used for numerical computations and efficient handling of array operations. Pandas was used for data manipulation, including loading, cleaning, filtering, and analyzing structured data. For data visualization, Matplotlib and Seaborn libraries were used to create various graphs and charts, helping in better understanding and interpretation of the data.

These tools and technologies collectively enabled efficient data processing, statistical analysis, feature engineering, and visualization, making it possible to extract meaningful insights from the dataset.

5 Methodology

The methodology of this project follows a systematic approach to analyze the dataset and extract meaningful insights. Initially, the dataset was collected and loaded into the Python environment using appropriate libraries. This was followed by data preprocessing, where irrelevant columns were removed, missing values were handled, and the dataset was prepared for analysis by selecting relevant numerical features.

After preprocessing, statistical analysis was performed to understand the distribution and characteristics of the data using measures such as mean, median, and standard deviation. Based on this, exploratory data analysis (EDA) was conducted to identify patterns, trends, and relationships between different variables such as social media usage, mental health, and academic performance.

Feature engineering techniques were applied to enhance the dataset by creating new features such as Risk Score, Age Group, and Usage Level, which helped in deeper analysis and better interpretation of the data. Finally, data visualization techniques were used to represent the findings through graphs and charts, enabling clear understanding of insights and supporting effective interpretation of results.

6 Implementation Details

Step 1: Data Loading

- The dataset was uploaded into the Google Colab environment for analysis.
- Required Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn were imported.
- The dataset was loaded using the Pandas library and stored in a DataFrame.
- The first five rows of the dataset were displayed to understand its structure and attributes.

Code:

```
from google.colab import files
uploaded = files.upload()

 # Import required libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load dataset
df = pd.read_excel("/content/student_social_media_analysis_101.xlsx")
# Display first 5 rows
df.head()
```

Output:

	Student_ID	Age	Gender	Academic_Level	Country	Avg_Daily_Usage_Hours	Most_Used_Platform
0	1	23	Male	High School	India	4.5	Facebook
1	2	20	Male	Undergraduate	UK	5.7	Instagram
2	3	24	Female	Undergraduate	Bangladesh	1.8	Instagram
3	4	21	Male	High School	Canada	7.7	Twitter
4	5	23	Male	Graduate	India	7.3	Facebook

Affects_Academic_Performance	Sleep_Hours_Per_Night	Mental_Health_Score	Conflicts_Over_Social_Media	Addicted_Score
Yes	5.3	5	2	4
Yes	6.0	3	3	6
No	6.4	8	3	6
No	7.0	6	1	8
No	6.0	7	2	1

Step 2: Data Preprocessing and Statistical Analysis

- Non-relevant columns such as Student ID were removed from the dataset.
- Only numerical columns were selected for analysis to focus on quantitative data.
- The cleaned dataset was displayed to verify preprocessing steps.
- Descriptive statistical analysis was performed on the dataset.
- Key measures such as mean, median, standard deviation, minimum, and maximum values were calculated.
- A statistical summary was generated to understand data distribution and variability.
- These measures helped in identifying patterns and preparing the data for further analysis.

Code:

```
# DATA PREPROCESSING
# Remove non-relevant column (if exists)
if "Student_ID" in df.columns:
    df = df.drop(columns=["Student_ID"])

# Keep only numeric columns
df_numeric = df.select_dtypes(include=['number'])

print("Cleaned Dataset:\n")
print(df_numeric.head())
# STATISTICAL ANALYSIS
```

```
print("\n--- Statistical Summary ---")
print(df_numeric.describe())

print("\n--- Mean Values ---")
print(df_numeric.mean())

print("\n--- Median Values ---")
print(df_numeric.median())

print("\n--- Standard Deviation ---")
print(df_numeric.std())

print("\n--- Minimum Values ---")
print(df_numeric.min())

print("\n--- Maximum Values ---")
print(df_numeric.max())
```

Output:

```
... Cleaned Dataset:

   Age  Avg_Daily_Usage_Hours  Sleep_Hours_Per_Night  Mental_Health_Score  \
0   23                    4.5                    5.3                    5
1   20                    5.7                    6.0                    3
2   24                    1.8                    6.4                    8
3   21                    7.7                    7.0                    6
4   23                    7.3                    6.0                    7

   Conflicts_Over_Social_Media  Addicted_Score
0                            2               4
1                            3               6
2                            3               6
3                            1               8
4                            2               1
```

```
--- Statistical Summary ---
              Age  Avg_Daily_Usage_Hours  Sleep_Hours_Per_Night  \
count  101.000000             101.000000             101.000000
mean    21.297030               5.043564               6.321782
std      2.636454               1.847345               1.098690
min     17.000000               1.600000               4.600000
25%     19.000000               3.700000               5.500000
50%     21.000000               5.200000               6.200000
75%     24.000000               6.700000               7.100000
max     25.000000               8.000000               8.500000

       Mental_Health_Score  Conflicts_Over_Social_Media  Addicted_Score
count           101.000000                   101.000000      101.000000
mean              6.267327                     2.326733        4.574257
std               2.014404                     1.738441        2.562602
min               3.000000                     0.000000        1.000000
25%               5.000000                     1.000000        2.000000
50%               7.000000                     2.000000        4.000000
75%               8.000000                     4.000000        7.000000
max               9.000000                     5.000000        9.000000
```

```
--- Mean Values ---
Age                              21.297030
Avg_Daily_Usage_Hours             5.043564
Sleep_Hours_Per_Night             6.321782
Mental_Health_Score               6.267327
Conflicts_Over_Social_Media       2.326733
Addicted_Score                    4.574257
dtype: float64
```

```
--- Median Values ---
Age                              21.0
Avg_Daily_Usage_Hours             5.2
Sleep_Hours_Per_Night             6.2
Mental_Health_Score               7.0
Conflicts_Over_Social_Media       2.0
Addicted_Score                    4.0
dtype: float64
```

```
--- Standard Deviation ---
Age                              2.636454
Avg_Daily_Usage_Hours            1.847345
Sleep_Hours_Per_Night            1.098690
Mental_Health_Score              2.014404
Conflicts_Over_Social_Media      1.738441
Addicted_Score                   2.562602
dtype: float64
```

```
--- Minimum Values ---
Age                              17.0
Avg_Daily_Usage_Hours             1.6
Sleep_Hours_Per_Night             4.6
Mental_Health_Score               3.0
Conflicts_Over_Social_Media       0.0
Addicted_Score                    1.0
dtype: float64
```

```
--- Maximum Values ---
Age                              25.0
Avg_Daily_Usage_Hours             8.0
Sleep_Hours_Per_Night             8.5
Mental_Health_Score               9.0
Conflicts_Over_Social_Media       5.0
Addicted_Score                    9.0
dtype: float64
```

Step 3: Handling Missing Values and Data Types

- The dataset was checked for missing values using appropriate functions.
- The number of missing values in each column was identified and displayed.
- Missing values (if present) were handled using mean imputation for numerical columns.
- This ensured that the dataset remained consistent and suitable for analysis.
- The data types of all columns were examined to understand the nature of each attribute.
- This step helped in verifying that the dataset was properly structured for further processing and analysis.

Code:

```
# Check missing values
print("\nMissing Values:")
print(df.isnull().sum())

# Fill missing values (if any)
df.fillna(df.mean(numeric_only=True), inplace=True)

# Data types
print("\nData Types:")
print(df.dtypes)
```

Output:

```
Missing Values:
Student_ID                      0
Age                             0
Gender                          0
Academic_Level                  0
Country                         0
Avg_Daily_Usage_Hours           0
Most_Used_Platform              0
Affects_Academic_Performance    0
Sleep_Hours_Per_Night           0
Mental_Health_Score             0
Conflicts_Over_Social_Media     0
Addicted_Score                  0
dtype: int64
```

```
Data Types:
Student_ID                        int64
Age                               int64
Gender                           object
Academic_Level                   object
Country                          object
Avg_Daily_Usage_Hours           float64
Most_Used_Platform               object
Affects_Academic_Performance     object
Sleep_Hours_Per_Night           float64
Mental_Health_Score               int64
Conflicts_Over_Social_Media       int64
Addicted_Score                    int64
dtype: object
```

Step 4: Country-wise Analysis and Risk Score Calculation

- The dataset was grouped based on the country attribute to perform comparative analysis.
- Mean values of average daily usage hours, addiction score, and mental health score were calculated for each country.
- The grouped data was sorted based on addiction score to identify countries with higher or lower average addiction levels.

- A custom feature called **Risk Score** was created using a weighted combination of average daily usage hours, addiction score, and mental health score.
- This metric helped in quantifying the overall risk associated with social media usage among students.
- Country-wise average risk scores were calculated to compare risk levels across different regions.
- The results were displayed to highlight variations in usage patterns and associated risks.
- A bar chart was created to visually compare addiction scores across countries.
- Proper labeling, grid lines, and value annotations were added to enhance readability and interpretation of the graph.

Code:

```
# Group by country
country_analysis = df.groupby('Country').agg({
    'Avg_Daily_Usage_Hours': 'mean',
    'Addicted_Score': 'mean',
    'Mental_Health_Score': 'mean'
})

# Sort by addiction score
country_analysis = country_analysis.sort_values(by='Addicted_Score')

print("\nCountry-wise Analysis:")
print(country_analysis)

# Create Risk Score (custom metric)
df['Risk_Score'] = (
    df['Avg_Daily_Usage_Hours'] * 0.4 +
    df['Addicted_Score'] * 0.3 +
    (10 - df['Mental_Health_Score']) * 0.3
)

# Country-wise risk
risk_country = df.groupby('Country')['Risk_Score'].mean().sort_values()

print("\nRisk Score by Country:")
```

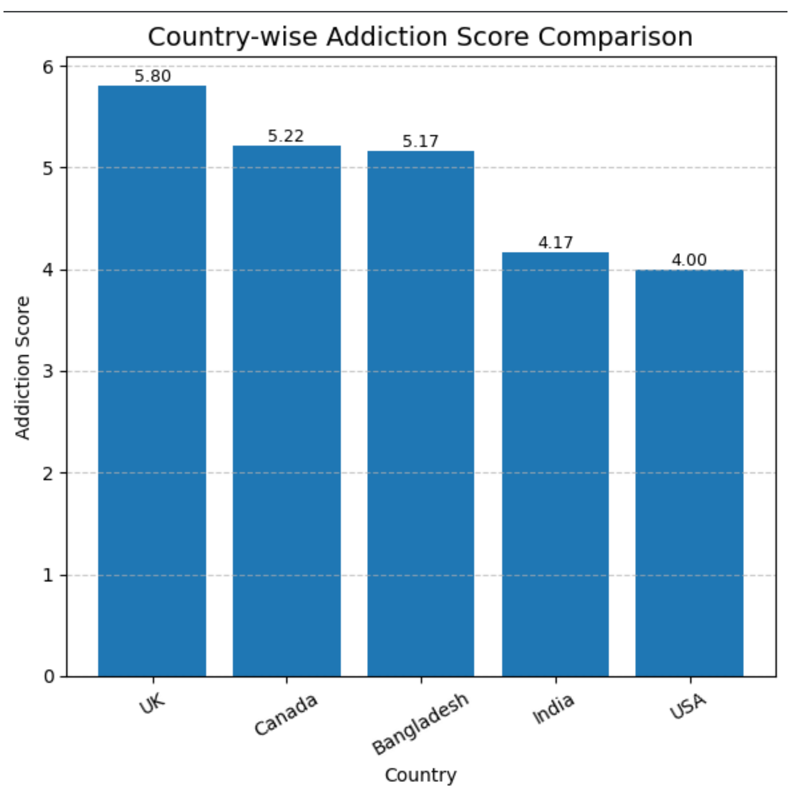
```
print(risk_country)

import matplotlib.pyplot as plt

# Sort data (important for clean look)
country_analysis   =   country_analysis.sort_values(by='Addicted_Score',
ascending=False)

plt.figure(figsize=(6,6))

bars = plt.bar(
    country_analysis.index,
    country_analysis['Addicted_Score']
)

# Titles
plt.title("Country-wise Addiction Score Comparison", fontsize=14)
plt.xlabel("Country")
plt.ylabel("Addiction Score")

# Rotate labels for readability
plt.xticks(rotation=30)

# Grid for better clarity
plt.grid(axis='y', linestyle='--', alpha=0.7)

# Add value labels on top of bars (VERY IMPORTANT)
for bar in bars:
    height = bar.get_height()
    plt.text(
        bar.get_x() + bar.get_width()/2,
        height,
        f'{height:.2f}',
        ha='center',
        va='bottom',
        fontsize=9
    )

plt.tight_layout()
plt.show()
```

Output:

```
Country-wise Analysis:
            Avg_Daily_Usage_Hours  Addicted_Score  Mental_Health_Score
Country
USA                      5.175000        4.000000             6.500000
India                    5.039583        4.166667             6.125000
Bangladesh               4.650000        5.166667             6.333333
Canada                   5.922222        5.222222             6.222222
UK                       4.770000        5.800000             6.500000
```

```
Risk Score by Country:
Country
USA           4.320000
India         4.428333
Bangladesh    4.510000
UK            4.698000
Canada        5.068889
Name: Risk_Score, dtype: float64
```

Graph:



Step 5: Age Group Analysis and Visualization

- The age attribute was categorized into groups such as Teen, Young Adult, Adult, and Mature using defined ranges.
- A new feature called **Age_Group** was created to simplify analysis based on age categories.
- The dataset was grouped according to age groups to analyze average daily usage hours, addiction scores, and mental health scores.
- This grouping helped in identifying behavioral patterns among different age categories.
- The number of students in each age group was calculated to understand the distribution of the dataset.
- A pie chart was created to visually represent the proportion of students in each age group.
- The visualization helped in clearly understanding the overall age distribution within the dataset.

Code:

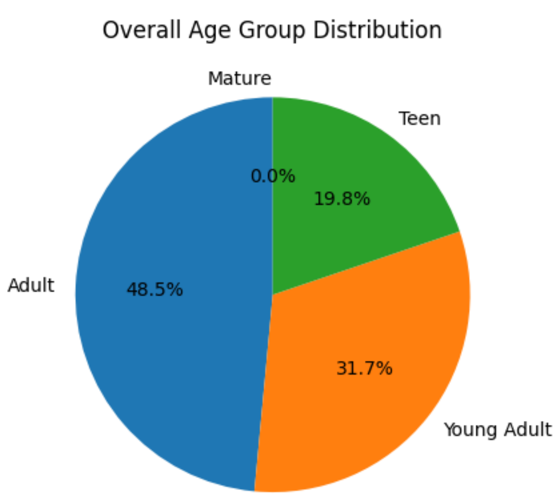
```
# Create age groups
bins = [15, 18, 21, 25, 30]
labels = ['Teen', 'Young Adult', 'Adult', 'Mature']

df['Age_Group'] = pd.cut(df['Age'], bins=bins, labels=labels)

# Group analysis
age_analysis = df.groupby('Age_Group').agg({
    'Avg_Daily_Usage_Hours': 'mean',
    'Addicted_Score': 'mean',
    'Mental_Health_Score': 'mean'
})

print("\nAge Group Analysis:")
print(age_analysis)
```

```
# Count number of people in each age group
age_counts = df['Age_Group'].value_counts()

# Plot pie chart
plt.figure()

plt.pie(
    age_counts,
    labels=age_counts.index,
    autopct='%1.1f%%',
    startangle=90
)

plt.title("Overall Age Group Distribution")
plt.show()
```

Output:

```
Age Group Analysis:
             Avg_Daily_Usage_Hours  Addicted_Score  Mental_Health_Score
Age_Group
Teen                      5.450000        4.500000             5.800000
Young Adult               5.015625        5.031250             6.531250
Adult                     4.895918        4.306122             6.285714
Mature                         NaN             NaN                  NaN
```

Graph:



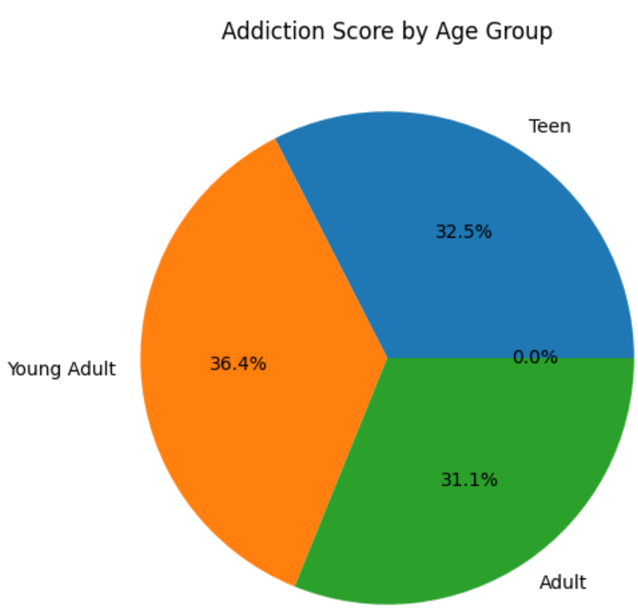
Step 6: Age Group-wise Comparative Visualization

- Pie charts were created to visualize the distribution of addiction scores across different age groups.
- This helped in understanding how social media addiction varies among different age categories.
- The use of percentage labels made the interpretation of data more intuitive and user-friendly.
- Titles and formatting were applied to ensure clarity and better presentation of the results.
- These visualizations helped in identifying patterns between age, addiction levels, and mental health conditions.

Code:

```
age_analysis['Addicted_Score'].plot(
    kind='pie',
    autopct='%1.1f%%',
    figsize=(6,6)
)
plt.title("Addiction Score by Age Group")
plt.ylabel("")
plt.show()
```

Output:



Step 7: Usage Level Classification

- The average daily usage hours were categorized into different levels such as Low, Medium, and High.
- A new feature called **Usage_Level** was created using defined ranges of screen time.
- This categorization helped in simplifying the analysis of social media usage patterns.
- The number of students in each usage category was calculated to understand distribution.
- The results provided insight into how many students fall under low, moderate, or high usage levels.
- This classification made it easier to relate usage intensity with other factors like mental health and academic performance.

Code:

```
# Categorize usage level
df['Usage_Level'] = pd.cut(
    df['Avg_Daily_Usage_Hours'],
    bins=[0, 3, 6, 10],
    labels=['Low', 'Medium', 'High']
)

# Show counts
print("\nUsage Level Distribution:")
print(df['Usage_Level'].value_counts())
```

Output:

```
Usage Level Distribution:
Usage_Level
Medium    51
High      31
Low       19
Name: count, dtype: int64
```

Step 8: Identification of High-Risk Students

- Students were filtered based on specific conditions to identify high-risk individuals.
- The criteria included students with high social media usage (more than 6 hours per day) and low mental health scores (less than 5).
- This filtering helped in isolating a group that may be negatively affected by excessive social media usage.
- Relevant attributes such as country, age, usage hours, and mental health score were displayed for these students.
- This step provided meaningful insights into vulnerable groups within the dataset.
- It also highlighted the potential risks associated with high screen time and poor mental well-being.

Code:

```
high_risk = df[
    (df['Avg_Daily_Usage_Hours'] > 6) &
    (df['Mental_Health_Score'] < 5)
]

print("\n🚨 High Risk Students:")
print(high_risk[['Country', 'Age', 'Avg_Daily_Usage_Hours',
'Mental_Health_Score']])
```

Output:

```
🚨 High Risk Students:
       Country  Age  Avg_Daily_Usage_Hours  Mental_Health_Score
21       India   22                    6.8                    3
23       India   17                    6.8                    4
52         USA   22                    7.1                    4
65  Bangladesh   21                    7.7                    3
66       India   24                    7.0                    3
80       India   19                    7.6                    3
```

Step 9: Ideal Student Guidelines and Healthy Group Identification

- Based on the analysis, ideal guidelines for student health and lifestyle were defined.
- Recommended screen time was identified as 2 to 3 hours per day to reduce addiction risk.
- Ideal sleep duration was set between 7 to 8 hours per night to support better cognitive and mental functioning.
- A mental health score range of 8 to 10 was considered as an indicator of good well-being.
- These guidelines were derived from observed patterns in the dataset.
- A subset of students meeting these criteria was filtered to identify "ideal healthy students."
- Conditions included low screen time, sufficient sleep, and high mental health scores.
- The total number of such students was calculated to understand how many meet the ideal conditions.
- Sample records of these students were displayed for better interpretation.
- This step demonstrated how data analysis can be used to provide practical recommendations for improving student lifestyle and well-being.

Code:

```
print("="*50)
print("        IDEAL STUDENT HEALTH GUIDELINES")
print("="*50)

print("\n1. Screen Time Recommendation:")
print("   - Optimal Usage: 2 to 3 hours per day")
print("   - Rationale: Minimizes digital fatigue and addiction risk")

print("\n2. Sleep Recommendation:")
```

```
print("   - Ideal Duration: 7 to 8 hours per night")
print("   - Minimum Requirement: 6 hours")
print("   - Rationale: Supports cognitive function and mental
stability")

print("\n3. Mental Health Benchmark:")
print("   - Desired Score: 8 to 10 (on a scale of 1–10)")
print("   - Interpretation: Indicates good psychological well-being")

print("\nConclusion:")
print("Maintaining balanced screen time along with adequate sleep")
print("contributes significantly to better mental health outcomes.")

print("="*50)
ideal_students = df[
    (df['Avg_Daily_Usage_Hours'] <= 3) &
    (df['Sleep_Hours_Per_Night'] >= 6) &
    (df['Mental_Health_Score'] >= 8)
]

print("Number of Ideal Healthy Students:", len(ideal_students))
ideal = ideal_students.head()

print("Ideal Students:")
print(ideal[['Avg_Daily_Usage_Hours', 'Sleep_Hours_Per_Night',
'Mental_Health_Score', 'Affects_Academic_Performance']])
```

Output:

```
==================================================
        IDEAL STUDENT HEALTH GUIDELINES
==================================================

1. Screen Time Recommendation:
   - Optimal Usage: 2 to 3 hours per day
   - Rationale: Minimizes digital fatigue and addiction risk

2. Sleep Recommendation:
   - Ideal Duration: 7 to 8 hours per night
   - Minimum Requirement: 6 hours
   - Rationale: Supports cognitive function and mental stability

3. Mental Health Benchmark:
   - Desired Score: 8 to 10 (on a scale of 1–10)
   - Interpretation: Indicates good psychological well-being

Conclusion:
Maintaining balanced screen time along with adequate sleep
contributes significantly to better mental health outcomes.
==================================================
Number of Ideal Healthy Students: 3
Ideal Students:
    Avg_Daily_Usage_Hours  Sleep_Hours_Per_Night  Mental_Health_Score  \
2                     1.8                    6.4                    8
27                    2.9                    7.7                    8
64                    1.8                    6.7                    8

   Affects_Academic_Performance
2                            No
27                           No
64                           No
```

7 Statistical Analysis

Statistical analysis was performed on the dataset to understand the distribution and behavior of key variables related to social media usage among students. Descriptive statistical measures such as mean, median, standard deviation, minimum, and maximum values were calculated for numerical attributes including average daily usage hours, sleep duration, mental health score, and addiction score. These measures helped in identifying the central tendency and variability within the dataset.

The statistical summary provided insights into how the data is distributed and highlighted differences among students in terms of their usage patterns and well-being indicators. It also helped in detecting any extreme values or inconsistencies in the dataset. This analysis served as a foundation for further exploratory data analysis by providing a clear understanding of the overall characteristics and trends present in the data.

8 Exploratory Data Analysis & Feature Engineering

Exploratory Data Analysis (EDA) was performed to identify patterns, trends, and relationships within the dataset related to students' social media usage. The analysis revealed noticeable relationships between average daily usage hours and factors such as sleep duration, mental health score, and academic performance. It was observed that higher social media usage is often associated with reduced sleep hours and a greater likelihood of negative impact on academic performance. Additionally, higher addiction scores showed a tendency to correspond with lower mental health scores, indicating a potential negative influence on students' well-being.

To enhance the analysis, feature engineering techniques were applied by creating new variables from existing data. A custom feature called Risk Score was developed using a weighted combination of average daily usage hours, addiction score, and mental health score to quantify the overall impact of social media usage. Furthermore, the Age attribute was categorized into groups such as Teen, Young Adult, Adult, and Mature to analyze behavioral differences across age segments. Similarly, social media usage was classified into Low, Medium, and High categories to better understand usage intensity.

Correlation analysis and grouping techniques were used to identify important features and relationships within the dataset. These steps improved the quality of analysis and provided deeper insights into how social media usage affects different aspects of students' lives.

9 Data Visualization and Interpretation

Data visualization techniques were used to present the analyzed data in a clear and meaningful manner. Various types of graphs such as bar charts and pie charts were created using Matplotlib and Seaborn to illustrate patterns and comparisons within the dataset. These visualizations helped in understanding the distribution of students across different categories and the relationship between key variables.

Bar charts were used to compare addiction scores across different countries, providing insights into regional variations in social media usage behavior. Pie charts were used to represent the distribution of age groups and to compare addiction and mental health scores across these groups. These visual representations made it easier to identify trends, such as higher usage levels being associated with increased addiction and lower mental health scores.

Overall, the visualizations supported the findings obtained from statistical analysis and exploratory data analysis. They enhanced the interpretation of data by presenting complex information in a simple and understandable format, thereby helping to draw meaningful conclusions about the impact of social media usage on students.

10 Results and Insights

The analysis of the dataset revealed several important insights regarding social media usage among students. It was observed that students with higher average daily social media usage are more likely to experience negative effects on their academic performance. Increased usage was also associated with reduced sleep duration, which can impact concentration and productivity.

The study further showed that higher addiction scores tend to correspond with lower mental health scores, indicating a negative relationship between excessive social media usage and psychological well-being. Country-wise analysis highlighted variations in addiction levels and risk scores, suggesting that social media behavior may differ across regions.

Age group analysis revealed that different age categories exhibit varying usage patterns, addiction levels, and mental health conditions. Additionally, students categorized under high usage levels were more likely to fall into the high-risk group, characterized by high screen time and lower mental health scores.

On the other hand, students who maintained lower usage levels, adequate sleep, and higher mental health scores were identified as “ideal healthy students.” These findings emphasize the importance of balanced social media usage and its positive impact on academic performance and overall well-being.

11 Conclusion

This project successfully demonstrated how data science techniques can be applied to analyze social media usage among students and its impact on academic performance, mental health, and daily habits. By using tools such as Python, NumPy, Pandas, and visualization libraries, meaningful patterns and relationships were identified within the dataset.

The analysis showed that excessive social media usage is associated with negative outcomes such as reduced sleep duration, lower mental health scores, and a greater impact on academic performance. On the other hand, students who maintained balanced usage, adequate sleep, and better mental health exhibited more stable academic behavior.

Overall, the project highlights the importance of maintaining a balanced digital lifestyle and demonstrates how data-driven approaches can be used to understand and improve student well-being. It also reflects the practical application of data science techniques in solving real-world problems related to student behavior and lifestyle.

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